Hope Enterprise University College	
Faculty of Informatics	
Department of Computer Science	
Course Outline	
Course Title:	**Design and Analysis of Algorithms**
Course Code:	**CS 442**
Credit Hours:	**3**
Contact Hours:	**4**
Lab Hours:	**2**
Prerequisite:	**Data Structures and Algorithm Analysis**
Course Category:	**Major**
Year:	**4**
Semester:	**1**
Instructor Name	**Belay Mengistu**

Course Description

The course focuses on the design and analysis of algorithms. Topics Include: Review of the basic data structures; Design techniques: divide-and-conquer, dynamic programming, greedy algorithms, And graph algorithms: Elementary graph algorithms, Breadth-first search (BFS), Depth-first search (DFS), Strongly-connected components, Minimum spanning tree, shortest paths.

Course Objectives

By the end of this course, students will be able to:

- Perform algorithm analysis using the different techniques;
- Demonstrate the use of algorithm design techniques; and
- Describe the basics of computational complexity
- Apply advanced searching and sorting algorithms
- Develop, and reason about the correctness and performance of algorithms, in particular for string
- Searching and graph manipulation

Course Contents	Weeks
Chapter 1: Introduction and Elementary Data Structures (6hr) 1.1. Introduction to Algorithm analysis	**1-2**

1.2. Review of elementary Data Structures	
Chapter 2: Divide and Conquer (6hr) 2.1. The General Method of Divide and Conquer 2.2. Binary Search 2.3. Finding Maximum and Minimum 2.4. Merge Sort 2.5. Quick Sort 2.6. Selection Sort	**3-5**
Chapter 3: Greedy Algorithms (6hr) 3.1. General Characteristic of Greedy Algorithms 3.2. Graph Minimum Spanning Tree (MST) - Kruskal's and Prims's Algorithms 3.3. Shortest Paths 3.4. Scheduling	**6-8**
Mid Exam	**9**
Chapter 4: Dynamic Programming (6hr) 4.1. Introduction to Dynamic Programming 4.2. All pairs Shortest Path - Floyd-Warshall Algorithm 4.3. Shortest Path - Dijkstra Algorithm 4.4. 0/1 Knapsack 4.5. Depth First Search	**10-12**
Chapter 5: Back Tracking (6hr) 5.1. Queens Problem 5.2. Graph Coloring 5.3. Hamiltonian Cycle 5.4. Knapsack Problems 5.5. Traveling Salesman Problems	**13-14**
Chapter 6: Introduction to Probabilistic and Parallel Algorithms (2hr) 6.1 Fundamentals of Probabilistic Algorithms 6.2 Fundamentals of Parallel Algorithms 6.3 Analysis of Probabilistic and Parallel Algorithms 6.4 Basic Design Techniques	**15 - 16**

Teaching and Learning Method: Lecture supported by lab session and class work exercises

Assessment /Evaluation Scheme

Component	Weight	Coverage	Assessment Time		
Class Work	5%	Chapter 1 & 2	Week 5		
Mid	**20%**	**Chapter 1, 2 & 3**	**Week 9**		
Test	15%	Chapter 4 & 5	Week 14		
Assignment	10%	Chapter 4, 5 & 6	Week 15		
Final Exam	**50%**	**All Chapters**			
Total	100%				

Text Book

- T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, Introduction to algorithms, third edition, MIT Press / McGraw-Hill, 2009.

Reference Books

- Algorithms for Dummies John Paul Mueller, Luca Massaron, Wiley, 2017
- Ammeraal. Leendert (1988). Programs and Data Structures in C (2nd ed.): John Wiley & Sons.
- Reingold, Edward M and Wilfred J. Hansen (1983) Data Structures: CBS Publishers & Distributors.
- Sahni, Sartaj (1999) Data Structures, Algorithms and Applications in C++: McGraw-Hill.
- Horowitz, Ellis, Sartaj Sahni and Dinesh Mehta (1995). Fundamentals of Data Structures in C++: Computer Science Press.